1 Expectation and Variance in Data Science and Machine Learning

Expectation and variance arise naturally in many data science and machine learning problems. We first consider their role in stochastic gradient descent, and then see how moments of a distribution can be used to estimate unknown parameters.

1.1 Expectation in Mini-batch Stochastic Gradient Descent

Consider a simple scalar linear regression model

$$\widehat{y} = wx,$$

where $w \in \mathbb{R}$ is the parameter to be learned.

Suppose our training data consist of the four observations

$$(1,0), \qquad (2,5), \qquad (3,6), \qquad (4,9).$$

These data approximately follow a linear relationship between x and y, but individual observations deviate from the fitted line, as is typical for noisy real-world data.

For a training example (x_i, y_i), use the squared loss

$$\ell_i(w) = \frac{1}{2}(wx_i - y_i)^2.$$

The gradient of the loss with respect to w is

$$\frac{d}{dw}\ell_i(w) = (wx_i - y_i)x_i.$$

Suppose that the current parameter value is

$$w = 2.$$

The gradient contributed by each of the four training examples is therefore

$$g_i = \frac{d}{dw}\ell_i(w)\Big|_{w=2} = (2x_i - y_i)x_i.$$

The corresponding predictions, residuals, and gradients are

(x_i, y_i)	$\widehat{y}_i = 2x_i$	$\widehat{y}_i - y_i$	$g_i = (2x_i - y_i)x_i$
$(1,0)$	2	2	2
$(2,5)$	4	-1	-2
$(3,6)$	6	0	0
$(4,9)$	8	-1	-4

If we use the entire dataset, the average gradient is

$$g_{\text{full}} = \frac{1}{4}(2 - 2 + 0 - 4) = -1.$$

Thus a full gradient descent update would use the direction

$$-1.$$

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Using one randomly selected training example

Instead of computing the gradient using all four training examples, suppose that we choose one example uniformly at random and use only its gradient.

Let

$$G = \text{gradient obtained from the randomly selected example.}$$

Then G has PMF

g	2	0	-2	-4
$P(G=g)$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$

Hence

$$\begin{aligned} E[G] &= 2\left(\frac{1}{4}\right) + 0\left(\frac{1}{4}\right) - 2\left(\frac{1}{4}\right) - 4\left(\frac{1}{4}\right) \\ &= -1. \end{aligned}$$

Therefore,

$$\boxed{E[G] = -1 = g_{\text{full}}.}$$

Thus, although the gradient obtained from a single randomly selected training example is random, its average value agrees exactly with the gradient computed using the full dataset.

This is one of the fundamental ideas behind stochastic gradient descent: a computationally inexpensive random gradient can still give the correct gradient direction on average.

Using a mini-batch of two training examples

Now suppose that we select two training examples independently and uniformly with replacement, and average their gradients.

Let the two sampled gradients be G_1 and G_2, and define

$$\overline{G} = \frac{G_1 + G_2}{2}.$$

Each sampled gradient can take one of the four values

$$2, \qquad 0, \qquad -2, \qquad -4.$$

Since each draw corresponds to choosing one of four training examples, there are

$$4 \times 4 = 16$$

equally likely ordered pairs.

For each pair, the average gradient is shown below:

$G_1 \backslash G_2$	2	0	-2	-4
2	2	1	0	-1
0	1	0	-1	-2
-2	0	-1	-2	-3
-4	-1	-2	-3	-4

Counting the number of times each value appears gives the PMF

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$\bar{g}$	2	1	0	−1	−2	−3	−4
$P(\overline{G}=\bar{g})$	$\frac{1}{16}$	$\frac{2}{16}$	$\frac{3}{16}$	$\frac{4}{16}$	$\frac{3}{16}$	$\frac{2}{16}$	$\frac{1}{16}$

We can calculate its expectation directly:

$$E[\overline{G}] = 2\left(\frac{1}{16}\right) + 1\left(\frac{2}{16}\right) + 0\left(\frac{3}{16}\right) - 1\left(\frac{4}{16}\right) - 2\left(\frac{3}{16}\right) - 3\left(\frac{2}{16}\right) - 4\left(\frac{1}{16}\right).$$

Thus

$$E[\overline{G}] = \frac{2+2-4-6-6-4}{16} = -1.$$

Therefore,

$$\boxed{E[\overline{G}] = E[G] = g_{\text{full}} = -1.}$$

Averaging two randomly sampled gradients has not changed the average gradient.

What happens to the variability?

For the one-sample gradient,

$$E[G^2] = 4\left(\frac{1}{4}\right) + 0\left(\frac{1}{4}\right) + 4\left(\frac{1}{4}\right) + 16\left(\frac{1}{4}\right) = 6.$$

Therefore,

$$\text{Var}(G) = E[G^2] - (E[G])^2 = 6 - (-1)^2 = 5.$$

For the two-sample average,

$$\begin{aligned} E[\overline{G}^2] &= 4\left(\frac{1}{16}\right) + 1\left(\frac{2}{16}\right) + 0\left(\frac{3}{16}\right) + 1\left(\frac{4}{16}\right) \\ &\quad + 4\left(\frac{3}{16}\right) + 9\left(\frac{2}{16}\right) + 16\left(\frac{1}{16}\right) \\ &= \frac{56}{16} \\ &= \frac{7}{2}. \end{aligned}$$

Hence

$$\text{Var}(\overline{G}) = E[\overline{G}^2] - (E[\overline{G}])^2 = \frac{7}{2} - 1 = \frac{5}{2}.$$

Thus

	Expectation	Variance
One sampled example G	−1	5
Average of two samples $\overline{G}$	−1	$\frac{5}{2}$

The mean gradient remains unchanged, but the mini-batch gradient has smaller variability.

Mini-batching preserves the average gradient while reducing its variability.

Notice that all of these calculations were obtained directly from the PMFs. We did not need to use general results about sums of independent random variables.

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2 Using Moments to Estimate Unknown Parameters

Expectation and variance are also useful for statistical estimation.

Suppose that the distribution of a random variable X depends on one or more unknown parameters. Population quantities such as

$$E[X], \qquad E[X^2], \qquad \mathrm{Var}(X)$$

often depend on these unknown parameters.

Given observed data

$$X_1, \ldots, X_m,$$

we can calculate the empirical mean

$$\overline{X} = \frac{1}{m} \sum_{i=1}^{m} X_i$$

and the empirical variance

$$V_m = \frac{1}{m} \sum_{i=1}^{m} (X_i - \overline{X})^2.$$

The basic idea of the *method of moments* is to replace unknown population moments by their observed empirical counterparts and solve for the unknown parameters.

2.1 Estimating the Support of a Uniform Distribution

Suppose

$$X \sim \mathrm{Unif}(0, \theta),$$

where the upper endpoint

$$\theta > 0$$

is unknown.

The density is

$$f_X(x) = \begin{cases} \dfrac{1}{\theta}, & 0 \le x \le \theta, \\ 0, & \text{otherwise.} \end{cases}$$

We will construct two estimators of the same unknown parameter θ: one based on the expectation and one based on the variance.

Estimator based on the expectation

We have

$$E[X] = \int_0^\theta x \frac{1}{\theta}\, dx.$$

Therefore,

$$\begin{aligned} E[X] &= \frac{1}{\theta} \left[\frac{x^2}{2} \right]_0^\theta \\ &= \frac{\theta}{2}. \end{aligned}$$

Thus

$$\theta = 2E[X].$$

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The population expectation is unknown, but the empirical mean $\overline{X}$ can be calculated from data. Replacing $E[X]$ by $\overline{X}$ gives

$$\boxed{\widehat{\theta}_{\text{mean}} = 2\overline{X}.}$$

Estimator based on the variance

First calculate

$$E[X^2] = \int_0^\theta x^2 \frac{1}{\theta}\, dx.$$

Hence

$$\begin{aligned} E[X^2] &= \frac{1}{\theta}\left[\frac{x^3}{3}\right]_0^\theta \\ &= \frac{\theta^2}{3}. \end{aligned}$$

Since

$$E[X] = \frac{\theta}{2},$$

we obtain

$$\begin{aligned} \operatorname{Var}(X) &= E[X^2] - (E[X])^2 \\ &= \frac{\theta^2}{3} - \frac{\theta^2}{4} \\ &= \frac{\theta^2}{12}. \end{aligned}$$

Therefore,

$$\theta = \sqrt{12 \operatorname{Var}(X)}.$$

Replacing the population variance by the empirical variance gives

$$\boxed{\widehat{\theta}_{\text{var}} = \sqrt{12 V_m}.}$$

Thus we have two different estimators:

$$\boxed{\widehat{\theta}_{\text{mean}} = 2\overline{X}, \qquad \widehat{\theta}_{\text{var}} = \sqrt{12 V_m}.}$$

Numerical example

Suppose we observe

$$1, \qquad 2, \qquad 8, \qquad 9.$$

Then

$$\overline{X} = \frac{1+2+8+9}{4} = 5.$$

The estimator based on the mean is therefore

$$\widehat{\theta}_{\text{mean}} = 2(5) = \boxed{10}.$$

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Now calculate the empirical variance:

$$\begin{aligned} V_4 &= \frac{1}{4}\left[(1-5)^2 + (2-5)^2 + (8-5)^2 + (9-5)^2\right] \\ &= \frac{1}{4}(16+9+9+16) \\ &= \frac{50}{4} \\ &= 12.5. \end{aligned}$$

The variance-based estimator is therefore

$$\begin{aligned} \widehat{\theta}_{\text{var}} &= \sqrt{12(12.5)} \\ &= \sqrt{150} \\ &\approx 12.25. \end{aligned}$$

Thus

$$\boxed{\widehat{\theta}_{\text{mean}} = 10, \qquad \widehat{\theta}_{\text{var}} \approx 12.25.}$$

The same observed data can therefore give different estimates depending on which population feature is matched.

This illustrates that expectation and variance are not merely descriptive summaries of a distribution. They can also contain information about unknown parameters of the underlying statistical model.

2.2 Estimating the Parameters of a Binomial Distribution

Suppose

$$X \sim \text{Bin}(n, p),$$

where both n and p are unknown.

For example, X may represent the number of failures in a collection of n similar trials, where p is the probability of failure on each trial. We observe only the total number of failures over many repetitions and would like to infer the underlying values of n and p.

For a binomial random variable,

$$E[X] = np$$

and

$$\text{Var}(X) = np(1-p).$$

Let

$$\mu = E[X], \qquad \sigma^2 = \text{Var}(X).$$

We therefore have the two equations

$$\mu = np$$

and

$$\sigma^2 = np(1-p).$$

Using $np = \mu$,

$$\sigma^2 = \mu(1-p).$$

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Hence

$$\frac{\sigma^2}{\mu} = 1 - p,$$

and therefore

$$\boxed{p = 1 - \frac{\sigma^2}{\mu}.}$$

Once p is known,

$$n = \frac{\mu}{p}.$$

Substituting the expression for p,

$$\begin{aligned} n &= \frac{\mu}{1 - \frac{\sigma^2}{\mu}} \\ &= \frac{\mu^2}{\mu - \sigma^2}. \end{aligned}$$

Thus

$$\boxed{n = \frac{\mu^2}{\mu - \sigma^2}.}$$

Replacing the population mean and variance by their empirical counterparts gives the method-of-moments estimators

$$\boxed{\widehat{p} = 1 - \frac{V_m}{\overline{X}}, \qquad \widehat{n} = \frac{\overline{X}^2}{\overline{X} - V_m}.}$$

Numerical example

Suppose that from $m = 10$ repetitions we observe

$$0, \quad 1, \quad 1, \quad 2, \quad 2, \quad 2, \quad 2, \quad 3, \quad 3, \quad 4.$$

The empirical mean is

$$\begin{aligned} \overline{X} &= \frac{0 + 1 + 1 + 2 + 2 + 2 + 2 + 3 + 3 + 4}{10} \\ &= \frac{20}{10} \\ &= 2. \end{aligned}$$

Now calculate the empirical variance:

$$\begin{aligned} V_{10} = \frac{1}{10}\big[&(0-2)^2 + (1-2)^2 + (1-2)^2 \\ &+ (2-2)^2 + (2-2)^2 + (2-2)^2 + (2-2)^2 \\ &+ (3-2)^2 + (3-2)^2 + (4-2)^2\big] \\ = \frac{1}{10}&\,[4 + 1 + 1 + 0 + 0 + 0 + 0 + 1 + 1 + 4] \\ = \frac{12}{10}& \\ = 1.2.& \end{aligned}$$

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Therefore,

$$\begin{aligned}\widehat{p} &= 1 - \frac{1.2}{2} \\ &= 1 - 0.6 \\ &= \boxed{0.4}.\end{aligned}$$

Similarly,

$$\begin{aligned}\widehat{n} &= \frac{2^2}{2 - 1.2} \\ &= \frac{4}{0.8} \\ &= \boxed{5}.\end{aligned}$$

Hence the method-of-moments estimates are

$$\boxed{\widehat{n} = 5, \qquad \widehat{p} = 0.4.}$$

We can verify that these estimates reproduce the empirical moments:

$$\widehat{n}\widehat{p} = 5(0.4) = 2 = \overline{X},$$

and

$$\widehat{n}\widehat{p}(1 - \widehat{p}) = 5(0.4)(0.6) = 1.2 = V_{10}.$$

Thus the first two moments of the observed data have allowed us to estimate two unknown parameters of the binomial distribution.

A note on estimating n

The parameter n of a binomial distribution must be an integer. In a general dataset, however, the method-of-moments estimate

$$\widehat{n} = \frac{\overline{X}^2}{\overline{X} - V_m}$$

need not be an integer.

Thus an algebraically obtained estimate may not automatically satisfy all constraints imposed by the statistical model.

Summary

The examples above illustrate two important uses of expectation and variance.

- In stochastic gradient descent, expectation describes the average behavior of a randomly selected gradient. Averaging multiple sampled gradients can preserve the same expected gradient while reducing variability.
- In statistical estimation, expectation and variance contain information about unknown parameters of a probability model. Matching population moments with empirical moments leads to the method of moments.

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